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Reprogrammable Mechanics via Individually Switchable Bistable Unit Cells in a Prestrained Chiral Metamaterial

*A. B. M. Tahidul Haque, Samuele Ferracin, and Jordan R. Raney**

Architected materials exhibit unique properties and functionalities based on the geometric arrangement of their constituent materials. In most cases, these parameters are fixed, requiring that the system be redesigned and reconstructed if different properties are desired. Both stimuli-responsive materials and modular designs have been used to enable re-programmable properties in the past, but often have limitations, such as the need for a continuous application of external stimuli or power, or unwanted global morphing. In this study, a locally stable anti-tetra chiral (LSAT) metamaterial is introduced consisting of independently multistable units that can deform and change state without inducing changes in the global morphology. Adjacent cells are only weakly coupled, allowing the collective metamaterial to be switched between many different possible states. Local bistability enables re-programmable heterogeneity, such as the snapping of cells along an edge or diagonally within the architected material. Utilizing finite element analysis (FEA), the influence of key geometric parameters on the re-programmability of the metamaterials is systematically investigated. The effect of these parameters on properties such as shear stiffness, Poisson's ratio, and vibration are also investigated using experimental prototypes. This re-programmable metamaterial promises to expand the design space for mechanical systems, with potential applications in non-traditional computation, robotic actuation, and adaptive structures.

1. Introduction

In recent decades, many mechanical metamaterials with exceptional properties have been developed, each of which derives its properties from subtle geometric features and material distribution. These advances have been enabled, in part, by breakthroughs in digital manufacturing and optimization,[1] allowing control of a wide variety of properties, including stiffness,[2] Poisson's ratio,[3] thermal expansion,[4] and morphing modes,[5] arising from meticulously designed geometric units and material distribution. While such architected materials have been used in various applications, including soft robotics,[6] deployable morphing structures,[7] flexible electronics,[8] and biomedical devices,[9] in most cases the properties of the metamaterial are fixed during synthesis. These structures could find broader applicability and adoption if their exceptional properties could be reprogrammed after synthesis.

Several approaches have been used to produce tunable metamaterials in the past. These include manipulation of boundaries,[10] the introduction of defects,[11] and the inclusion of hierarchical features.[12] Other approaches include controllable topological interlocking,[13] multistable morphing,[14] and geometric frustration.[15] Jamming of loose granular particles,[16] interlocked granular elements,[13] and layered mediums[17] can be used to achieve adjustable stiffness and shape. However, jamming-based re-programmable metamaterials commonly need complex pneumatic leakage-proof enclosures and external vacuum pumps to operate. The miniaturization of these systems is also complex because of the intrinsic size of the granular materials used. Buckling-induced multistability is another frequently studied approach that is used to design metamaterials with programmable force-displacement relationships, Poisson's ratio, or other properties, but this approach often necessitates global morphing or volumetric changes.[14,18]

Here, we design a mechanical metamaterial that allows control of properties via digits that can be independently switched between multiple distinct states, without significantly influencing the state of adjacent unit cells. This allows for numerous combinations of switched patterns, generating a larger and more complex design space. We describe a locally stable anti-tetrachiral (LSAT) system, composed of independently and reversibly switchable chiral digits. By inducing an asymmetrical bistability via prestrain, the system serves as a re-programmable metamaterial, enabling local tunability without volumetric changes, in contrast with prior work.[4,19–21] We study

A. B. M. Tahidul Haque, S. Ferracin, J. R. Raney
Department of Mechanical Engineering and Applied Mechanics
University of Pennsylvania
Philadelphia, PA 19104, USA
E-mail: raney@seas.upenn.edu

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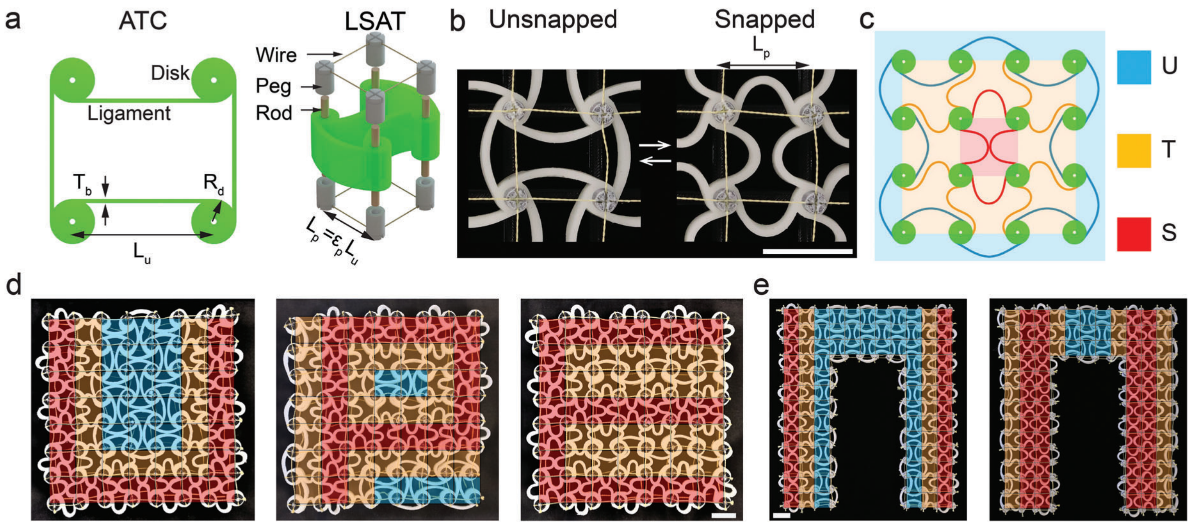

Figure 1. Locally stable mechanical digits. a) A regular ATC cell showing the critical geometric parameters (left). Exploded view (right) shows that a regular ATC cell can be prestrained via wire frames to produce the LSAT cell. b) Experimental images of the two stable states of a prestrained cell; each cell can be reversibly snapped between these states. c) A schematic of a 3 × 3 LSAT, showing how changing the state of the center cell (i.e., from unsnapped, 'U', to snapped, 'S', shown in red) produces a transition region (see orange ligaments, 'T') in the immediately adjacent cells, but has no effect on the cells beyond these (see unsnapped ligaments, 'U', in blue). d) Reversible snapping of an LSAT in the pattern of the letters U, P, and E. e) An inverted U-shaped structure with two different configurations of the cells. All scale bars are 2 cm.

the snapping mechanism and its influence on global mechanical properties, including shear modulus, Poisson's ratio, and propagation of linear waves. By achieving tunable properties without volumetric change, the LSAT structure presented here could contribute to multifunctional applications involving sensors, actuators, damping, and mechanical computers. Responsive or active materials could be incorporated with the LSAT structure using approaches from 4D printing[22,23] and soft robotics[24] to enable autonomous switching and interactive reprogramming.

2. Concept and Design

In this study, we start with a regular anti-tetrachiral (ATC) system.[4,19–21] A single ATC unit consists of four disks connected by four ligaments, as shown in **Figure 1a**, left. The unit length, disk radius, and beam thickness are denoted L_u, R_d, and T_b, respectively. Experimental specimens of the ATC are created via mold casting, using a silicone (Dragon Skin 30) with glass fibers (10 vol.%). First, the higher stiffness of the fiber-reinforced Dragon Skin helps prevent out-of-plane buckling when specimens are oriented vertically during tests. Second, using a material with a higher stiffness results in higher mechanical loads during the experiments, reducing the relative measurement uncertainty that results from the intrinsic error of the load cell. The thickness of the specimen is chosen to avoid out-of-plane deformation, while still being easy to extract from the mold during the fabrication process. As-fabricated, the ATC is monostable. If the beams of a cell are deformed, they will return to their initial state when the external load is removed. This regular, monostable ATC is made into a bistable LSAT system by applying prestrain using a wire frame (Figure 1a, right), which pre-compresses and bends the beams. This prestrain is achieved by fitting two identical wire frames on opposite sides of an LSAT structure. The frame is made of equally spaced pegs connected by tensioned wires (see details in Experimental Section). The prestrain boundary condition forces each disk to reposition itself so that the distance between adjacent disks becomes equal to the pre-strained length $L_p(= (1 - \varepsilon_p)L_u)$. Here, L_u is the length of the relaxed beams and ε_p is the applied prestrain. Once prestrained, the beams can be reversibly snapped between two stable states, referred to as unsnapped and snapped (Figure 1b). In a larger LSAT structure with many unit cells, the state change of one unit cell remains localized, only affecting the nearest neighbors, which deform into an S-shaped configuration. These S-shaped unit cells constitute a transition region, serving as an interface between completely snapped and unsnapped cells, decoupling their deformation fields. Figure 1c illustrates the mechanism of a locally switchable metamaterial, where the snapped cell at the center (red) is surrounded by transition cells (orange), leaving the ligaments on the border unsnapped (blue). Additionally, the limited coupling between elements enables various formations of switched cells in the same LSAT structure. Video S1 (Supporting Information) shows an example of the bistable units of an experimental LSAT specimen being manually snapped between states. Figure 1d shows some illustrative switchable configurations, in this case with distributions that resemble various letters. Snapping a unit cell does not change the distance between disks; consequently, the global shape is not affected by the distribution of unsnapped and snapped cells. This is a significant deviation from traditional snapping architectures, where a change of state is usually accompanied by structural reconfiguration.[25–27] Moreover, there is no requirement that the LSAT be in the shape of a square. For example, the LSAT could be designed in other shapes such as the inverted U of Figure 1e, which may impart hierarchical mechanical properties. Regardless of the global architecture, the individual cells remain reversibly switchable. Moreover, different spatial distributions of snapped units are associated with different

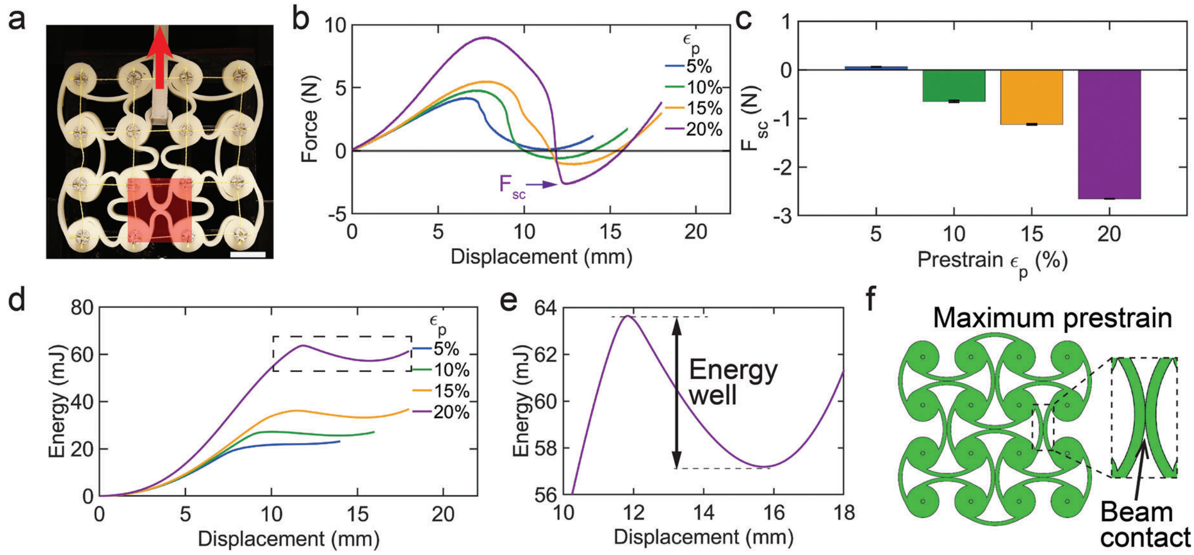

Figure 2. Effect of prestrain ϵ_p. a) Experimental image showing a specimen being subjected to a force, as indicated by the blue arrow. The red shaded cell is pre-snapped to keep the center cell partially snapped before loading. The scale bar indicates 2 cm. b) Force-displacement relationship for the loading scenario indicated in (a) as a function of prestrain ϵ_p. c) Second critical force F_{sc} as a function of ϵ_p. d) Potential energy versus displacement for different values of prestrain ϵ_p. e) Magnified view of the second energy well for the prestrain case $\epsilon_p = 0.20$, corresponding to the dashed box in (d). f) Upper prestrain limit ($\epsilon_p = 0.25$) of an LSAT structure with $R_d/L_u = 0.24$.

mechanical properties. Hence, switching the unit cells in heterogeneous patterns can potentially allow the macroscopic behavior of the metamaterials to be reversibly changed. In principle, the transition between unsnapped and snapped configurations could be achieved by a variety of soft actuators.[28] Pneumatic mechanisms, for example, can easily provide the amount of force and displacement necessary to snap a bistable element.[29] Responsive materials, such as hydrogels or liquid crystal elastomers (LCEs) can also be introduced to trigger state changes in bistable mechanisms.[30,31]

In the following sections we describe the modeling and experimental procedures used to characterize the reprogrammable LSAT metamaterials. First, we discuss our experimental study of the local bistability of the metamaterial as a function of the prestrain field. This determines the smallest amount of prestrain that is needed to make the units bistable. This information is then leveraged to develop a finite element (FE) model, enabling us to perform a numeric parametric study over the critical geometric parameters. The parametric study allows targeted design of LSAT structures that exhibit desired mechanical properties, ultimately allowing construction of a large LSAT structure used to experimentally validate the numerical predictions, specifically toward understanding the reprogrammability of shear stiffness, Poisson's ratio, etc. Finally, in addition to the static responses, we show that the LSAT structures can achieve tunable dynamic responses, e.g., for harmonic and impact excitation.

3. Bistable Chiral Cell as a Switchable Unit

In order for the ATC units to be bistable, a prestrain must be applied. Here, we experimentally investigate the bistability of an LSAT cell as a function of the prestrain that is applied by the wire frames. To quantify this effect, we vary the prestrain ϵ_p by fitting the LSAT system in different sized frames. More specifically, the snapping behavior is evaluated by measuring the second critical force F_{sc} (see **Figure 2**b) of the central cell of a 3 × 3 LSAT specimen ($R_d/L_u = 0.24$ and $T_b/L_u = 0.06$), as shown in Figure 2a. When F_{sc} is negative, the cell is bistable. The more negative the value of F_{sc}, the more stable is the second equilibrium. However, reliably measuring F_{sc} for the entire cell by simultaneously snapping all four beams is challenging. Therefore, we pre-snap the adjacent cell (in the red shaded box of Figure 2a), which partially snaps the central cell. This method allows us to complete the snapping process by applying force to just one beam (arrow in Figure 2a). Using this method, the force-displacement relations are measured for each prestrain ϵ_p, which is varied from 5% to 20% in 5% increments (by changing the wire frame), as shown in Figure 2b. F_{sc} is then extracted for each experiment. The average values are displayed in Figure 2c. As shown, for $\epsilon_p \leq 5\%$ the cells are not bistable. For $10\% \leq \epsilon_p \leq 20\%$ the absolute values of F_{sc} increase with ϵ_p. The bistability of the LSAT structure is associated with a double-well potential energy function (see Figure 2d). Figure 2e shows a magnified view of the second energy well for the prestrain case $\epsilon_p = 0.2$, indicated by the box in Figure 2d. The energy barrier between equilibria increases with ϵ_p, as shown in Figure S1a (Supporting Information). The effect of ϵ_p on the energy well is further elucidated via the analysis of a different LSAT structure with $R_d/L_u = 0.2$. Figure S2 (Supporting Information) shows that the energy barrier increases as a function of ϵ_p, and, therefore, that the cell becomes more stable in the second equilibrium state. However, if ϵ_p is too high, the beams may contact each other in the unsnapped configuration. For example, for the configuration $R_d/L_u = 0.24$, the beams come into contact when a prestrain $\epsilon_p = 0.25$ is applied (Figure 2f). Contact causes stiffening that affects the reprogrammability of the system. The prestrain ϵ_p should therefore remain between

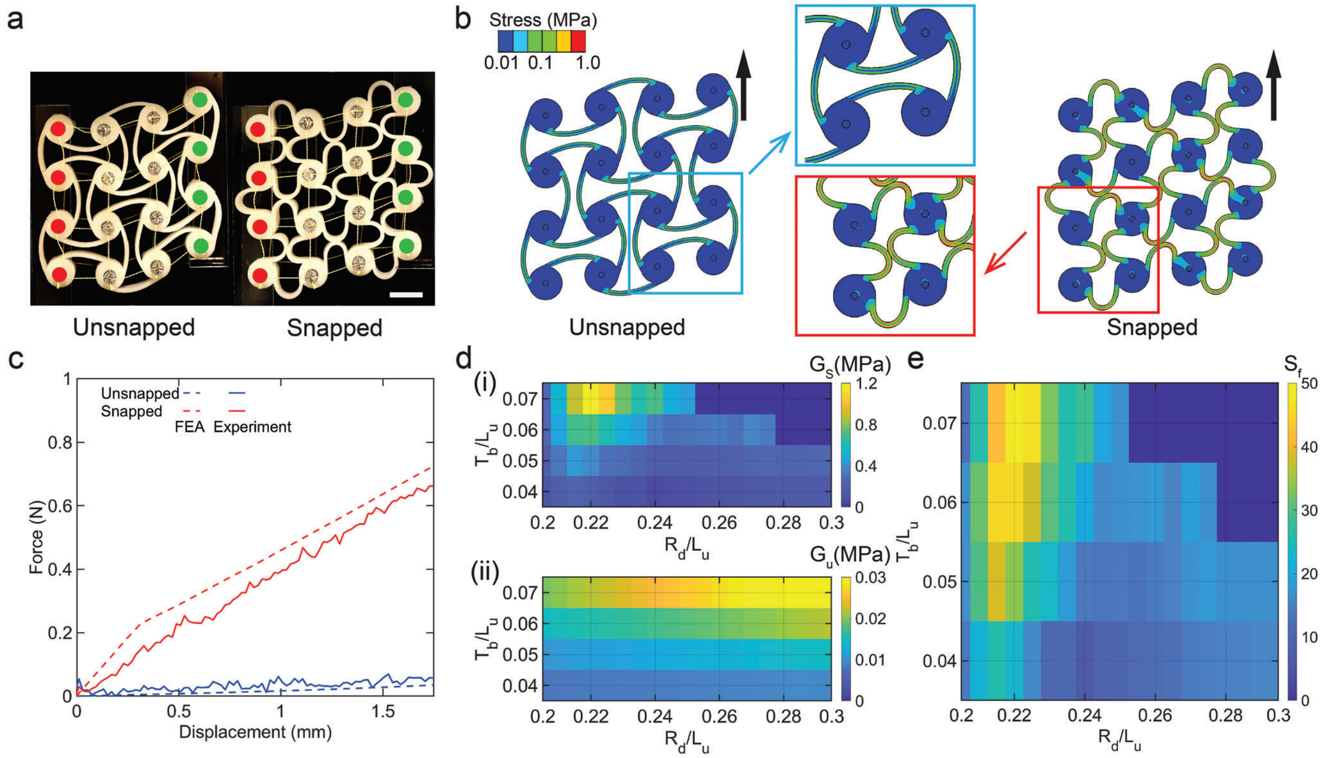

Figure 3. Parametric design for tunable properties. a) Shear loading of a 3 × 3 LSAT for snapped and unsnapped configurations. The disks marked by red and green circles indicate hinged and loaded disks, respectively. The blue arrows indicate loading edge. The scale bars are 2 cm. b) Stress distribution during shear loading, simulated in FEA for both configurations. Magnified views of the stress distribution of corner cells are shown in the middle. c) Shear force versus displacement for unsnapped (blue) and snapped (red) systems, obtained from FEA and experiments. d) Shear stiffness (indicated by color) for (i) snapped and (ii) unsnapped configurations as a function of T_b/L_u and R_d/L_u. e) Stiffness factor $S_f = G_s/G_u$.

0.05 (the onset of bistability) and 0.25 (the onset of self-contact). This prestrain limit depends on the geometry of the LSAT. For instance, for other LSAT geometric ratios such as $R_d/L_u = 0.20$ and 0.3, Figure S1b,c (Supporting Information) shows that the upper prestrain limit is 0.29 and 0.22, respectively. For the remainder of this study, we choose $\epsilon_p = 0.20$, which is the highest prestrain value that can be applied to the different structures analyzed in the parametric study of the next section ($0.2 \leqslant R_d/L_u \leqslant 0.3$).

4. Structural Design by Parametric Analysis

The previous section confirms that snapping is confined to individual cells, and it is decoupled from global morphing. LSAT systems can therefore embody distinct combinations of switched cells, each of which, as discussed later, can have distinct mechanical properties. In this section, we discuss a parametric study conducted via finite element analysis (FEA) that aims to maximize the reprogrammability of the shear stiffness between a snapped and unsnapped configuration.

As an example of this process, we start by studying the response of the LSAT structure under shear loading, with $T_b/L_u = 0.06$ and $R_d/L_u = 0.24$ (Figure 2) at $\epsilon_p = 0.2$. The experimental results for this design, for both completely unsnapped and snapped cases, are shown in **Figure** 3a (left and right, respectively). Shear loading is applied, as indicated by the blue arrows, to the four rightmost nodes (green), while the leftmost nodes (red) are held fixed via hinges. These tests have also been performed numerically using ABAQUS Standard (see Experimental Section). The experimental and numerical results are compared in Figure 3c (solid and dashed, respectively). Both results show a significant and consistent increase in the shear stiffness when the cells are put in the snapped configuration (the experimental data show a 20x increase, from 0.017 to 0.347 N mm^{-1}). The mechanism causing this stiffness increase is shown in the magnified view of Figure 3b. The buckled beams of the snapped cell (red box) come into contact, compress, and slide against each other, providing additional resistance. In contrast, beams of unsnapped cells (blue box) remain free from any contact and frictional interactions.

To further guide the design process, we use FEA to quantify the effect of key parameters. Figure 3d(i), (ii) show shear moduli as a function of T_b/L_u and R_d/L_u for the snapped (G_s) and unsnapped (G_u) configurations, respectively. The dark blue area of Figure 3d(i) and Figure 3e corresponding to $R_d/L_u > 0.28$, $T_b/L_u > 0.06$, and $R_d/L_u > 0.255$, $T_b/L_u > 0.07$ indicates structures for which cells cannot be switched due to geometric constraints. The ratios between the shear moduli of a snapped and unsnapped configuration (i.e., stiffness factor $S_f = G_s/G_u$) are displayed in Figure 3e. These results indicate a stiffness magnification of over 40x in the range of $0.05 < T_b/L_u < 0.07$ and $0.21 < R_d/L_u < 0.23$, with a peak of $S_f > 50$.

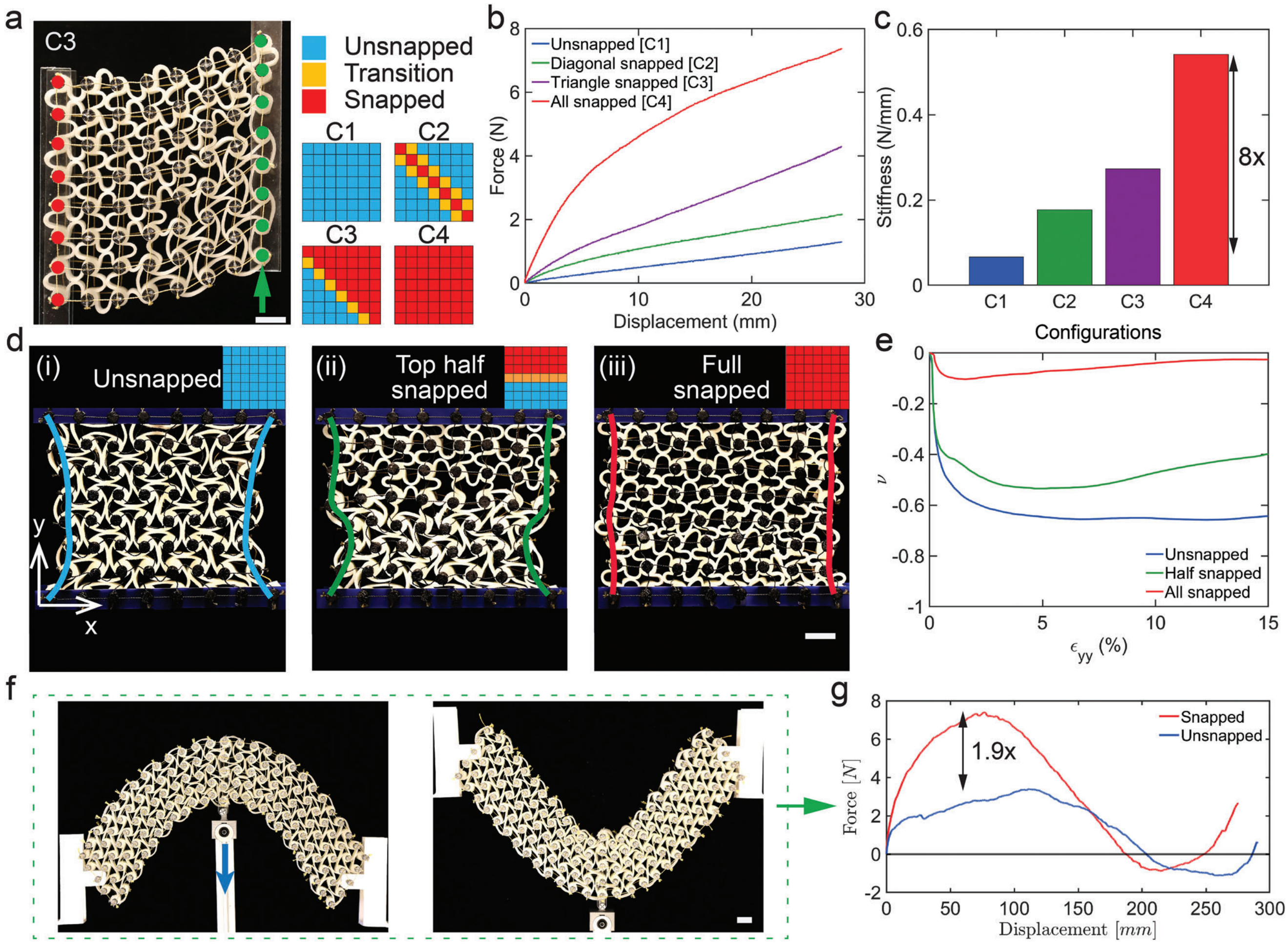

Figure 4. Reprogrammable mechanical response. a) Shear loading experiments on a 7×7 LSAT in configuration C3. The disks marked by red and green circles indicate supported and loaded disks, respectively. Switching patterns for four reversible configurations C1, C2, C3, and C4 are shown on the right side. b) Force-displacement relationships for different configurations of an LSAT with $\epsilon_p = 0.2$, obtained via shear experiments. c) Shear stiffness extracted from the experiments in (b). d) Auxetic experiments on (i) unsnapped, (ii) completely snapped, and (iii) half-snapped configurations at $\epsilon_{yy} = 15\%$. e) Poisson's ratio ν as a function of global strain ϵ_{yy} for three configurations. f) A hierarchical curved beam composed of LSAT cells can be snapped between two distinct buckled configurations by applying a force at the beam's center (blue arrow). g) The force-displacement relationship corresponding to a force applied at the blue arrow in (f). The load capacity of the beam is a factor of 1.9 higher when the cells are snapped relative to the unsnapped case. All scale bars are 2 cm.

4.1. Reprogrammable Mechanical Response

Above, we have only considered the properties of systems that are entirely snapped or entirely unsnapped. However, one of the potential benefits of the LSAT system is that many different configurations can be achieved by combining different arrangements of individually snapped and unsnapped cells. Next, we quantify the effect of different LSAT configurations on mechanical properties including shear stiffness and Poisson's ratio. We choose to study a 7×7 system with $R_d/L_u = 0.22$, $T_b/L_u = 0.07$. This system has a high G_s/G_u ratio (Figure 3e). As shown in **Figure 4**a, the four configurations studied are all unsnapped (C1), diagonally snapped (C2), triangularly snapped (C3), and completely snapped (C4). Here, blue, orange, and red colors indicate unsnapped, transition, and snapped configurations of the associated cells, respectively. Figure 4a (left) shows an experimental image of the triangularly snapped (C3) configuration being subjected to shear. Additional configurations are shown in Figure S3 (Supporting Information). Figure 4b shows the force-displacement relationships as experimentally measured via shear loading, for different configurations. The shear stiffness, values vary from 0.10 N mm^{-1} for configuration C1 to 0.50 N mm^{-1} for configuration C4 (Figure 4c). Similarly, the bar chart of Figure S4 (Supporting Information) shows a maximum force ranging between 2.6–6.6 N.

Next, we study the auxetic behavior of the same LSAT system. While the Poisson's ratio of a regular chiral metamaterial can only be tuned by a geometric redesign,[32,33] an LSAT structure's auxetic behavior can be reprogrammed based on the switched cells, as shown in Figure 4d. The Poisson's ratio ν is reported in Figure 4e as a function of global strain ϵ_{yy}, for the three configurations. The unsnapped LSAT of Figure 4d(i) behaves as a typical chiral system, reaching $\nu = -0.6$ at $\epsilon_{yy} = 2\%$; notably, the Poisson's ratio remains relatively constant from $\epsilon_{yy} = 5\%$ to 15%. In the case of the completely snapped structure of Figure 4d(ii),

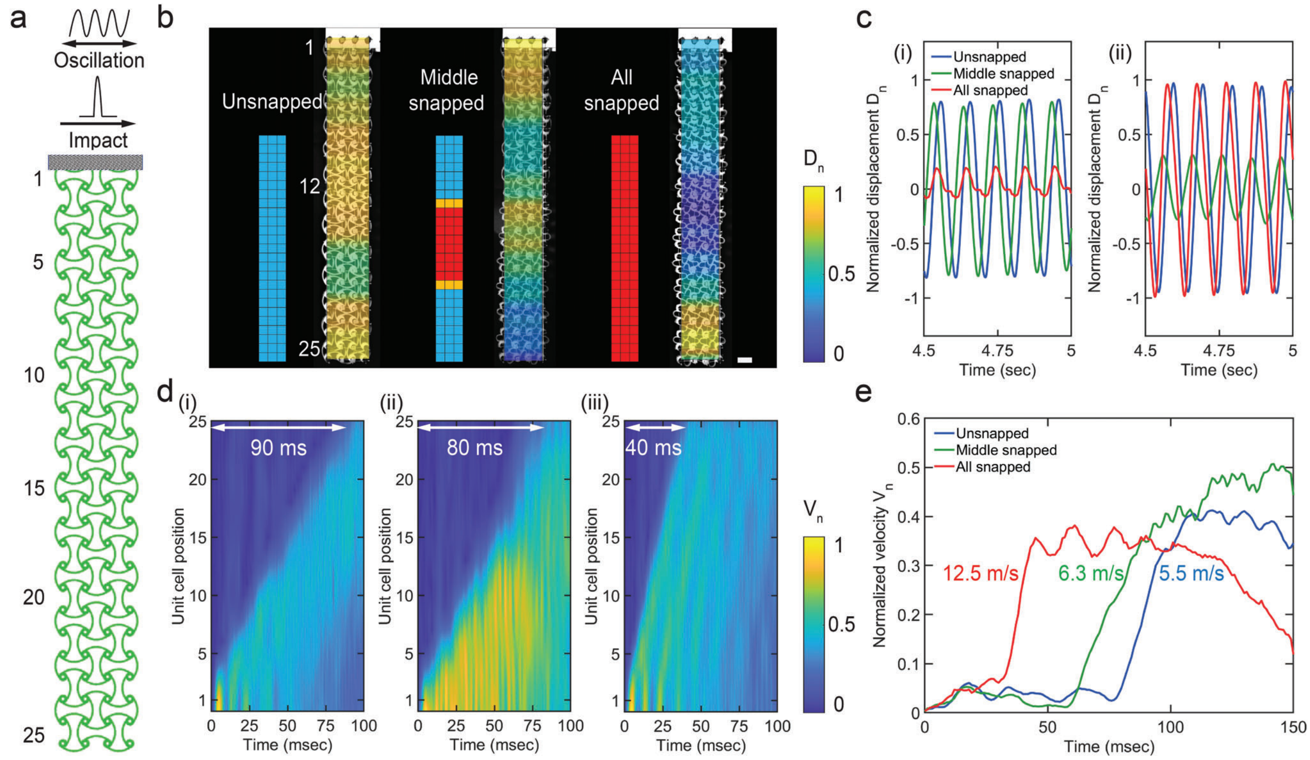

Figure 5. Dynamic responses of an LSAT structure. a) Schematic of a 25 × 3 LSAT structure which is hung vertically. Different dynamic loads can be applied, such as a harmonic oscillation or an impulse load. In either case, the load is applied at the top chiral disks, in shear with respect to the hanging structure. b) Experimental images with false color added indicating the normalized displacements D_n when a LSAT structure in three different configurations is subjected to harmonic loading. The scale bar is 2 cm. c) Normalized displacement D_n of the (i) 12th and (ii) 25th row for the three configurations. d) Color plot of the normalized velocity versus time for (i) completely unsnapped, (ii) middle snapped, and (ii) completely snapped configurations after the structures are subjected to an impulse at the top. e) Normalized velocities V_n of the 25th row as a function of time, for the three configurations. Propagation speeds $W_s = 12.5$ m/s, $W_{ms} = 6.3$ m/s, and $W_u = 5.5$ m/s are also reported next to the corresponding rise in oscillation velocities V_n.

the auxetic response is significantly different, with ν never lower than −0.01. The source of this tunable auxetic response is the cell-locking behavior of the snapped units, presented in the previous section. Interestingly, the LSAT can also behave heterogeneously in the mixed configuration of Figure 4d(iii), where the snapped and unsnapped segments, respectively, attain $\nu = -0.5$ and −0.1, resulting in an average value of −0.3. The blue, green, and red lines of Figure 4d connecting the external nodes are drawn to highlight the contrast in auxetic deformation for the different configurations.

The beams in a snapped cell come into contact, resulting in both a higher shear stiffness and a higher Poisson's ratio. The stiffness and Poisson's ratio of the collective metamaterial can therefore be tuned by changing the number of cells that are in the snapped state. In such a metamaterial, the snapped and unsnapped configurations give the upper and lower bounds, respectively, for these two properties, representing the extreme cases. Every other configuration will therefore have stiffness or Poisson's ratio between these two values. As a result, a configuration with increased stiffness also shows an increased Poisson's ratio.

Additionally, LSAT systems can be used to create hierarchical structures like beams. Here, we test the reprogrammability of a 26 × 4 bistable beam-shaped LSAT structure, as shown in Figure 4f. The beam is axially pre-compressed and the two ends are hinged. This allows the hierarchical beam to be globally bistable for both the unsnapped and completely snapped configurations. However, by switching all the units, we can significantly alter its force-displacement relationship, as indicated in Figure 4g. Specifically, the load capacity of the fully snapped beam can be increased by 90%. The reprogrammability of hierarchical structures can be exploited to create more complex systems in which some, or all, the components can be reprogrammed.

These three studies confirm that it is both conceptually and practically feasible to create a metamaterial that can be reprogrammed by reversible variations in the state of its unit cells.

4.2. Dynamic Response

The dynamic response of LSAT structures can also be reprogrammed via the configuration of snapped and unsnapped cells. Because shear stiffness is found to change significantly due to the snapped locking mechanism, a transverse excitation is used to excite the structure. More specifically, we focus on tuning the vibration modes and the transverse-wave propagation speed of a 25 × 3 LSAT structure. This is oriented vertically (as shown in **Figure 5a**) to avoid out-of-plane deformation under gravitational load and frictional interactions with any surfaces.

At first, the modes of oscillation for three different configurations are examined experimentally (Figure 5b). The horizontal harmonic excitation is applied to the top row by a shaker, which can precisely control the waveform, frequency, and amplitude. As expected, we find that the modes vary significantly for different configurations. Figure 5b presents the measured modes of transverse oscillation at 10 Hz by showing the normalized displacement $D_n = D_i/D_{max}$, where D_i is the displacement of a disk in ith row and D_{max} is the maximum amplitude in the system. The unsnapped case presents a combination of higher modes, which result in three high amplitude zones (at the top, middle, and bottom). The middle snapped configuration shows an oscillation mode strongly characterized by the high stiffness region at the center, which oscillates rigidly with a pendulum-like motion. Finally, the fully snapped configuration presents a different pendulum-like oscillation, in which the energy is mainly concentrated at the bottom of the system. Figure 5c highlights the difference in the normalized displacement D_n measured experimentally for these three configurations at the 12th (Figure 5c(i)) and 25th row (Figure 5c (ii)) of the LSAT structure. In this way, one can tune the modes of oscillation of an LSAT structure simply by changing which cells are snapped and unsnapped. Another example of tunable oscillation is presented in Figure S5a (Supporting Information), where the rows of LSAT cells are kept alternately snapped and unsnapped with a transition row between them. Here, we notice that D_n is well distributed throughout the structure during oscillation.

We also considered transverse nonlinear waves generated by a lateral impact. The horizontal impact is applied to the top row by a pendulum, and the propagation speed is measured using digital image correlation (DIC) on a sequence of images extracted from high-speed camera video (see Experimental Section). The 2D plots in Figure 5d(i), (ii), and (iii) display the normalized horizontal velocity $V_n = V_i/V_{max}$ (where V_i is the velocity of the ith row as a function of time, and V_{max} is the maximum velocity achieved in the system at any time), for unsnapped, middle snapped, and fully snapped configurations. The waves propagate from the first cell, at time $t = 0$ s, with substantially different speeds W, based on the switching composition of the structure. In the unsnapped case, with a lower shear stiffness, the propagation requires 90 ms to reach the end of the specimen (i.e., the 25th cell), while it takes 40 ms for the snapped case and 80 ms for the middle snapped case. This difference is also shown in Figure 5e, where it's possible to see the sudden increase in the normalized velocities V_n of the last row at different times. The propagation speed along the specimen is calculated to be $W_s = 12.5$ m/s, $W_{ms} = 6.3$ m/s, and $W_u = 5.5$ m/s for the snapped, middle snapped, and unsnapped configurations, respectively, i.e., a factor of 2.25x difference between snapped and unsnapped. Additionally, the wave propagation behavior of a fourth configuration is reported in Figure S5b (Supporting Information), where it takes nearly 75 ms for the wave to reach the last cell.

4.3. Conclusion

The LSAT architecture provides a strategy for rapid, local, and reversible tuning of mechanical properties without volumetric or shape changes. We have shown the effect of prestrain, which transforms a regular chiral system into an LSAT structure, both numerically and experimentally. The wire frames produce prestrain while still allowing global deformation. As a result, both static and dynamic properties, including the shear stiffness, auxetic response, and response to vibrations and impact loading, can be altered by reversibly snapping or unsnapping different units. Prestrained chiral cells can also be arranged in hierarchical structures with different shapes (e.g., rectangles, U-shapes, etc.). As the underlying snap-through mechanism of each unit is scale-invariant, miniaturization of LSAT systems is also possible. Although the current study deliberately does not consider the use of stimuli-responsive materials, in order to focus on the mechanical properties, it's possible that in the future responsive materials could be integrated with LSAT structures to allow autonomous property changes based on environmental stimuli (heat, light, solvents, etc.), as has been for other bistable mechanisms.[34,35]

5. Experimental Section

Material Characterization: Chiral specimens were molded from a composite material consisting of 0.8 mm glass microfibers (Fibre Glast) dispersed at 10 vol.%. in silicone elastomer (Dragon Skin 00-30, Smooth-On). The mechanical behavior of the composite was determined by uniaxial tension tests using an Instron 68SC-5 universal testing machine, following ASTM D412 guidelines. The measured stress–strain relationship was used to calibrate the nonlinear Yeoh hyperelastic model[36] by fitting the experimental data to the strain energy function:

$$U = \sum_{j=1}^{3} \left[C_{j0}[\bar{I}_1 - 3]^j + \frac{1}{D_j}[J - 1]^{2j} \right] \tag{1}$$

where, $J = det(\mathbf{F})$, $\bar{I}_1 = J^{-\frac{2}{3}tr(F^T F)}$, and $\mathbf{F}$ are the volumetric deformation, first invariant, and deformation gradient, respectively. The hyperelastic coefficients $C_{10} = 0.046$ MPa, $C_{20} = -0.28$ MPa, $C_{30} = 01.31$ MPa and $D_1 = D_2 = D_3 = 0$ were then used in the FE analysis material module. Figure S6 (Supporting Information) shows the Yeoh model and the experimental data as solid blue and dotted red lines, respectively.

LSAT Fabrication: The LSAT main body was fabricated using a mold casting method. The liquid phase composite was cast in a 3D printed mold (which was printed using a Fortus 450mc, Stratasys). The two wire frames that impose the prestrain boundary conditions were built by connecting 3D printed pegs with kevlar wire using super glue. The spacing between the pegs was maintained by placing them in a plate with equidistant holes during the gluing process. After this process, the cured chiral structure was placed between the frames; the molded part was kept in a prestrained position by a series of acrylic rods that pierce through the chiral disks and align the frames. This method readily prestrains the chiral structure to transform it into an LSAT system.

Curved Beam Snapping: The multistability of the curved beam shown in Figure 4f was investigated using an LSAT chain consisting of 26 × 4 cells. The chain was bent into an arc-like shape by two nodes located at the shorter edges using a 3D printed (MakerGear M2) hinge support. The curved hierarchical beam was positioned horizontally on a flat surface while the disks along the middle-line were attached to a T-slot aluminum rod that applied the load. The snap-through behavior was characterized in displacement control by pulling the aluminum rod at 10 mms^{-1} with a linear actuator (Velmex BiSlide). A load cell (LUX-B-50N-ID-P, Kyowa Electronic Instruments) was placed between the rod and the actuator to measure the force during the test.

FE Model Development and Validation: The finite element model was developed using ABAQUS Standard. CPE8H (full integration quadratic order hybrid) solid elements were used with an approximate global mesh size ranging from 0.25 to 0.8 mm. The previously described hyperelastic Yeoh constitutive model was implemented to describe the mechanical properties of the material. The critical set of parameters that affects the analysis is the friction involved during contact between beams (self-contact) and between the disks and the rods connected to the frame.

The 2D simulation involved four static steps. First, a prestrain was applied to the structure via the rods at the center of the LSAT's disks, allowing the structure to be locally switchable. Second, all the required unit cells were snapped; this was achieved by pulling the center of all the beams of each cell. After this, the structure was allowed to relax without any additional applied load. The relaxation step was necessary in the simulations to allow the beams to relax, as occurs experimentally after a beam is snapped and the force was removed. Finally, the required displacement was applied to the four rods on one edge to shear the structure, while the four rods on the opposite side were kept fixed, but free to rotate.

Auxetic Analysis: The geometry optimized for shear loading was also used to study the auxetic behavior. The structure was vertically compressed in the previously reported Instron machine and the in-plane deformation was monitored using a digital camera (Canon EOS 5D Mark IV). The videos were then analyzed by a custom Python DIC (Digital Image Correlation) code. Each disk was equipped with a tracker, allowing measurement of the displacement of each cell.

To calculate the Poisson's ratio of this discrete metamaterial, the method described by Overvelde et al.[37] was used. During each test, the local values of the engineering strain $\epsilon_{xx}^{[i,j]}$, and $\epsilon_{yy}^{[i,j]}$ were calculated as

$$\begin{aligned}\epsilon_{xx}^{[i,j]} &= \frac{x^{[i+1,j]} - x^{[i,j]} + x^{[i+1,j+1]} - x^{[i,j+1]} - 2L_0}{2L_0}\\ \epsilon_{yy}^{[i,j]} &= \frac{y^{[i+1,j]} - y^{[i,j]} + y^{[i+1,j+1]} - y^{[i,j+1]} - 2L_0}{2L_0}\end{aligned} \tag{2}$$

where i, $j = 1,\ldots, 8$ are the indexes representing each disk of the structure, and L_0 is the initial distance between the disks. The local values of the strain were then averaged ($\overline{\epsilon}_{xx}^{[i,j]}$, $\overline{\epsilon}_{yy}^{[i,j]}$) and used to calculate the global Poisson's ratio as

$$\nu^{[i,j]} = -\frac{\overline{\epsilon}_{xx}^{[i,j]}}{\overline{\epsilon}_{yy}^{[i,j]}} \tag{3}$$

Dynamic Tests: An LSAT comprising 25×3 units was molded to enable experimental characterization of the dynamic properties of LSAT structures. The structure was hung vertically from the top three unit cells, attached to an APS 113 Electro-Seis shaker. A harmonic waveform was imposed via a FeelTech FY2300 wave generator, whose signal was amplified with an APS 125 amplifier. The impact excitation was generated by a custom impact pendulum that collided at the top support where the shaker was attached. The same DIC software and trackers reported in the Auxetic Analysis section were used, but a high-speed camera (Photron FASTCAM Mini AX) was utilized to capture displacements at high-speed (1000 fps).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

chiral, metamaterial, multistable, physical intelligence, re-programmable

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